```
!pip install kaggle ultralytics opencv-python pandas matplotlib

!kaggle datasets download -d meowmeowmeowmeowmeow/gtsrb-german-traffic-sign
```

```
Dataset URL: https://www.kaggle.com/datasets/meowmeowmeowmeowmeow/gtsrb-german-traffic-sign
License(s): CC0-1.0
Downloading gtsrb-german-traffic-sign.zip to /content
100% 612M/612M [00:07<00:00, 90.0MB/s]
```

```
!unzip gtsrb-german-traffic-sign.zip

import os

train_path = "/content/Train"

print(os.listdir(train_path)[:5])
```

```
['23', '28', '4', '29', '0']
```

```
import pandas as pd

df = pd.read_csv("/content/Test.csv")
df.head()
```


```
{"summary":"{\n  \"name\": \"df\",\n  \"rows\": 12630,\n  \"fields\": [\n  {\n  \"column\": \"Width\",\n  \"properties\": {\n  \"dtype\": \"number\",\n  \"std\": 25,\n  \"min\": 25,\n  \"max\": 266,\n  \"num_unique_values\": 173,\n  \"samples\": [\n  160,\n  47,\n  112\n  ],\n  \"semantic_type\": \"\",\n  \"description\": \"\"\n  }\n  },\n  {\n  \"column\": \"Height\",\n  \"properties\": {\n  \"dtype\": \"number\",\n  \"std\": 23,\n  \"min\": 25,\n  \"max\": 232,\n  \"num_unique_values\": 163,\n  \"samples\": [\n  217,\n  161,\n  113\n  ],\n  \"semantic_type\": \"\",\n  \"description\": \"\"\n  }\n  },\n  {\n  \"column\": \"Roi.X1\",\n  \"properties\": {\n  \"dtype\": \"number\",\n  \"std\": 1,\n  \"min\": 1,\n  \"max\": 23,\n  \"num_unique_values\": 19,\n  \"samples\": [\n  6,\n  8,\n  16\n  ],\n  \"semantic_type\": \"\",\n  \"description\": \"\"\n  }\n  },\n  {\n  \"column\": \"Roi.Y1\",\n  \"properties\": {\n  \"dtype\": \"number\",\n  \"std\": 1,\n  \"min\": 5,\n  \"max\": 19,\n  \"num_unique_values\": 15,\n  \"samples\": [\n  16,\n  17,\n  5\n  ],\n  \"semantic_type\": \"\",\n  \"description\": \"\"\n  }\n  },\n  {\n  \"column\": \"Roi.X2\",\n  \"properties\": {\n  \"dtype\": \"number\",\n
```

\"std\": 23,\n   \"min\": 20,\n   \"max\": 244,\n
\"num_unique_values\": 167,\n   \"samples\": [\n   83,\n
139,\n   119\n   ],\n   \"semantic_type\": \"\",\n
\"description\": \"\"\n   }\n   },\n   {\n   \"column\":
\"Roi.Y2\",\n   \"properties\": {\n   \"dtype\": \"number\",\n
\"std\": 22,\n   \"min\": 20,\n   \"max\": 212,\n
\"num_unique_values\": 156,\n   \"samples\": [\n   116,\n
103,\n   72\n   ],\n   \"semantic_type\": \"\",\n
\"description\": \"\"\n   }\n   },\n   {\n   \"column\":
\"ClassId\",\n   \"properties\": {\n   \"dtype\": \"number\",\
n   \"std\": 11,\n   \"min\": 0,\n   \"max\": 42,\n
\"num_unique_values\": 43,\n   \"samples\": [\n   31,\n
30,\n   24\n   ],\n   \"semantic_type\": \"\",\n
\"description\": \"\"\n   }\n   },\n   {\n   \"column\":
\"Path\",\n   \"properties\": {\n   \"dtype\": \"string\",\n
\"num_unique_values\": 12630,\n   \"samples\": [\n
\"Test/08904.png\",\n   \"Test/11635.png\",\n
\"Test/04308.png\"\n   ],\n   \"semantic_type\": \"\",\n
\"description\": \"\"\n   }\n   }\n]\
n}","type":"dataframe","variable_name":"df"}

```python
import os
import cv2
import pandas as pd

base_path = "/content"
output_path = "/content/dataset"

os.makedirs(f"{output_path}/images/train", exist_ok=True)
os.makedirs(f"{output_path}/images/val", exist_ok=True)
os.makedirs(f"{output_path}/labels/train", exist_ok=True)
os.makedirs(f"{output_path}/labels/val", exist_ok=True)

df = pd.read_csv("/content/Test.csv")

from sklearn.model_selection import train_test_split
train_df, val_df = train_test_split(df, test_size=0.2)

def convert(df_split, split):
    for _, row in df_split.iterrows():
        img_path = os.path.join(base_path, row['Path'])

        if not os.path.exists(img_path):
            continue

        img = cv2.imread(img_path)
        h, w, _ = img.shape

        x_center = ((row['Roi.X1'] + row['Roi.X2']) / 2) / w
        y_center = ((row['Roi.Y1'] + row['Roi.Y2']) / 2) / h
```

```
        width = (row['Roi.X2'] - row['Roi.X1']) / w
        height = (row['Roi.Y2'] - row['Roi.Y1']) / h

        img_name = os.path.basename(img_path)

        cv2.imwrite(f"{output_path}/images/{split}/{img_name}", img)

        with
open(f"{output_path}/labels/{split}/{img_name.replace('.png','.txt')}"
, "w") as f:
            f.write(f"{int(row['ClassId'])} {x_center} {y_center}
{width} {height}")

convert(train_df, "train")
convert(val_df, "val")

print("□ Converted successfully!")
```

```
□ Converted successfully!
```

```
yaml_content = """path: /content/dataset
train: images/train
val: images/val

nc: 43
names:
['0','1','2','3','4','5','6','7','8','9','10','11','12','13','14','15'
,'16','17','18','19',

'20','21','22','23','24','25','26','27','28','29','30','31','32','33',
'34','35',
        '36','37','38','39','40','41','42']
"""

with open("/content/traffic.yaml", "w") as f:
    f.write(yaml_content)

print("□ YAML fixed!")
```

```
□ YAML fixed!
```

```
!pip install ultralytics

from ultralytics import YOLO

model = YOLO("yolov5s.pt")

model.train(
    data="/content/traffic.yaml",
    epochs=30,
    imgsz=416,
```

```
    batch=16
)

Creating new Ultralytics Settings v0.0.6 file 
View Ultralytics Settings with 'yolo settings' or at
'/root/.config/Ultralytics/settings.json'
Update Settings with 'yolo settings key=value', i.e. 'yolo settings
runs_dir=path/to/dir'. For help see
https://docs.ultralytics.com/quickstart/#ultralytics-settings.
PRO TIP  Replace 'model=yolov5s.pt' with new 'model=yolov5su.pt'.
YOLOv5 'u' models are trained with
https://github.com/ultralytics/ultralytics and feature improved
performance vs standard YOLOv5 models trained with
https://github.com/ultralytics/yolov5.

/ultralytics/assets/releases/download/v8.4.0/yolov5su.pt to
'yolov5su.pt': 100% ━━━━━━━━━━━━━━━━ 17.7MB 129.3MB/s 0.1s
Ultralytics 8.4.30  Python-3.12.13 torch-2.10.0+cu128 CUDA:0 (Tesla
T4, 14913MiB)
engine/trainer: agnostic_nms=False, amp=True, angle=1.0,
augment=False, auto_augment=randaugment, batch=16, bgr=0.0, box=7.5,
cache=False, cfg=None, classes=None, close_mosaic=10, cls=0.5,
compile=False, conf=None, copy_paste=0.0, copy_paste_mode=flip,
cos_lr=False, cutmix=0.0, data=/content/traffic.yaml, degrees=0.0,
deterministic=True, device=None, dfl=1.5, dnn=False, dropout=0.0,
dynamic=False, embed=None, end2end=None, epochs=30, erasing=0.4,
exist_ok=False, fliplr=0.5, flipud=0.0, format=torchscript,
fraction=1.0, freeze=None, half=False, hsv_h=0.015, hsv_s=0.7,
hsv_v=0.4, imgsz=416, int8=False, iou=0.7, keras=False, kobj=1.0,
line_width=None, lr0=0.01, lrf=0.01, mask_ratio=4, max_det=300,
mixup=0.0, mode=train, model=yolov5s.pt, momentum=0.937, mosaic=1.0,
multi_scale=0.0, name=train, nbs=64, nms=False, opset=None,
optimize=False, optimizer=auto, overlap_mask=True, patience=100,
perspective=0.0, plots=True, pose=12.0, pretrained=True,
profile=False, project=None, rect=False, resume=False,
retina_masks=False, rle=1.0, save=True, save_conf=False,
save_crop=False, save_dir=/content/runs/detect/train,
save_frames=False, save_json=False, save_period=-1, save_txt=False,
scale=0.5, seed=0, shear=0.0, show=False, show_boxes=True,
show_conf=True, show_labels=True, simplify=True, single_cls=False,
source=None, split=val, stream_buffer=False, task=detect, time=None,
tracker=botsort.yaml, translate=0.1, val=True, verbose=True,
vid_stride=1, visualize=False, warmup_bias_lr=0.1, warmup_epochs=3.0,
warmup_momentum=0.8, weight_decay=0.0005, workers=8, workspace=None
/assets/Arial.ttf to '/root/.config/Ultralytics/Arial.ttf': 100%
━━━━━━━━━━━━━━━━ 755.1KB 28.7MB/s 0.0s
Overriding model.yaml nc=80 with nc=43

                   from  n    params  module
arguments
```

```
  0                  -1  1      3520  ultralytics.nn.modules.conv.Conv
[3, 32, 6, 2, 2]
  1                  -1  1     18560  ultralytics.nn.modules.conv.Conv
[32, 64, 3, 2]
  2                  -1  1     18816  ultralytics.nn.modules.block.C3
[64, 64, 1]
  3                  -1  1     73984  ultralytics.nn.modules.conv.Conv
[64, 128, 3, 2]
  4                  -1  2    115712  ultralytics.nn.modules.block.C3
[128, 128, 2]
  5                  -1  1    295424  ultralytics.nn.modules.conv.Conv
[128, 256, 3, 2]
  6                  -1  3    625152  ultralytics.nn.modules.block.C3
[256, 256, 3]
  7                  -1  1   1180672  ultralytics.nn.modules.conv.Conv
[256, 512, 3, 2]
  8                  -1  1   1182720  ultralytics.nn.modules.block.C3
[512, 512, 1]
  9                  -1  1    656896
ultralytics.nn.modules.block.SPPF            [512, 512, 5]

 10                  -1  1    131584  ultralytics.nn.modules.conv.Conv
[512, 256, 1, 1]
 11                  -1  1         0
torch.nn.modules.upsampling.Upsample         [None, 2, 'nearest']

 12             [-1, 6]  1         0
ultralytics.nn.modules.conv.Concat           [1]

 13                  -1  1    361984  ultralytics.nn.modules.block.C3
[512, 256, 1, False]
 14                  -1  1     33024  ultralytics.nn.modules.conv.Conv
[256, 128, 1, 1]
 15                  -1  1         0
torch.nn.modules.upsampling.Upsample         [None, 2, 'nearest']

 16             [-1, 4]  1         0
ultralytics.nn.modules.conv.Concat           [1]

 17                  -1  1     90880  ultralytics.nn.modules.block.C3
[256, 128, 1, False]
 18                  -1  1    147712  ultralytics.nn.modules.conv.Conv
[128, 128, 3, 2]
 19            [-1, 14]  1         0
ultralytics.nn.modules.conv.Concat           [1]

 20                  -1  1    296448  ultralytics.nn.modules.block.C3
[256, 256, 1, False]
 21                  -1  1    590336  ultralytics.nn.modules.conv.Conv
[256, 256, 3, 2]
```

```
 22            [-1, 10]  1         0
ultralytics.nn.modules.conv.Concat           [1]

 23                  -1  1   1182720  ultralytics.nn.modules.block.C3
[512, 512, 1, False]
 24        [17, 20, 23]  1   2132689
ultralytics.nn.modules.head.Detect           [43, 16, None, [128, 256,
512]]
YOLOv5s summary: 154 layers, 9,138,833 parameters, 9,138,817
gradients, 24.1 GFLOPs

Transferred 421/427 items from pretrained weights
Freezing layer 'model.24.dfl.conv.weight'
AMP: running Automatic Mixed Precision (AMP) checks...
/ultralytics/assets/releases/download/v8.4.0/yolo26n.pt to
'yolo26n.pt': 100% ━━━━━━━━━━━━ 5.3MB 95.3MB/s 0.1s
AMP: checks passed 
train: Fast image access  (ping: 0.0±0.0 ms, read: 174.4±40.9 MB/s,
size: 6.0 KB)
train: Scanning /content/dataset/labels/train... 10104 images, 0
backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 10104/10104 2.5Kit/s 4.0s
train: New cache created: /content/dataset/labels/train.cache
albumentations: Blur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01,
blur_limit=(3, 7)), ToGray(p=0.01, method='weighted_average',
num_output_channels=3), CLAHE(p=0.01, clip_limit=(1.0, 4.0),
tile_grid_size=(8, 8))
val: Fast image access  (ping: 0.0±0.0 ms, read: 361.4±261.4 MB/s,
size: 15.1 KB)
val: Scanning /content/dataset/labels/val... 2526 images, 0
backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 2526/2526 1.5Kit/s 1.7s
val: New cache created: /content/dataset/labels/val.cache
optimizer: 'optimizer=auto' found, ignoring 'lr0=0.01' and
'momentum=0.937' and determining best 'optimizer', 'lr0' and
'momentum' automatically...
optimizer: AdamW(lr=0.000213, momentum=0.9) with parameter groups 69
weight(decay=0.0), 76 weight(decay=0.0005), 75 bias(decay=0.0)
Plotting labels to /content/runs/detect/train/labels.jpg...
Image sizes 416 train, 416 val
Using 2 dataloader workers
Logging results to /content/runs/detect/train
Starting training for 30 epochs...

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
Size
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━ 79/79 5.2it/s 15.3s
                 all       2526       2526      0.697      0.614
0.712       0.61

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
```

```
Size
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━ 79/79 6.1it/s 12.9s
                   all       2526       2526      0.893      0.834
0.917      0.789

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
Size
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━ 79/79 6.2it/s 12.8s
                   all       2526       2526      0.895      0.895
0.956      0.844

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
Size
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━ 79/79 6.3it/s 12.6s
                   all       2526       2526       0.94      0.948
0.978      0.828

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
Size
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━ 79/79 6.2it/s 12.7s
                   all       2526       2526       0.94      0.971
0.99      0.857

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
Size
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━ 79/79 6.2it/s 12.7s
                   all       2526       2526      0.967      0.979
0.989      0.868

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
Size
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━ 79/79 5.3it/s 14.8s
                   all       2526       2526      0.969      0.981
0.991      0.908

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
Size
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━ 79/79 6.2it/s 12.7s
                   all       2526       2526      0.981      0.981
0.994      0.918

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
Size
```

```
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━ 79/79 6.2it/s 12.8s
                   all       2526       2526      0.985      0.987
0.993      0.909

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
Size
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━ 79/79 6.3it/s 12.6s
                   all       2526       2526      0.985      0.989
0.994      0.909

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
Size
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━ 79/79 6.3it/s 12.6s
                   all       2526       2526      0.989      0.991
0.993      0.904

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
Size
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━ 79/79 6.3it/s 12.5s
                   all       2526       2526      0.992      0.993
0.994      0.898

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
Size
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━ 79/79 6.2it/s 12.6s
                   all       2526       2526      0.989      0.992
0.993      0.931

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
Size
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━ 79/79 6.2it/s 12.7s
                   all       2526       2526      0.992      0.995
0.995      0.936

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
Size
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━ 79/79 6.4it/s 12.4s
                   all       2526       2526      0.993      0.995
0.995      0.928

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
Size
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
```

```
━━━━━━━━━━━━ 79/79 6.2it/s 12.8s
                   all       2526       2526      0.993      0.996
0.994      0.929

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
Size
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━ 79/79 6.3it/s 12.6s
                   all       2526       2526      0.993      0.994
0.995      0.941

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
Size
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━ 79/79 6.2it/s 12.7s
                   all       2526       2526      0.993      0.997
0.994      0.943

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
Size
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━ 79/79 6.4it/s 12.3s
                   all       2526       2526      0.992      0.996
0.993      0.935

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
Size
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━ 79/79 6.3it/s 12.5s
                   all       2526       2526      0.994      0.996
0.993      0.943
Closing dataloader mosaic
albumentations: Blur(p=0.01, blur_limit=(3, 7)), MedianBlur(p=0.01,
blur_limit=(3, 7)), ToGray(p=0.01, method='weighted_average',
num_output_channels=3), CLAHE(p=0.01, clip_limit=(1.0, 4.0),
tile_grid_size=(8, 8))

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
Size
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━ 79/79 6.3it/s 12.5s
                   all       2526       2526      0.994      0.997
0.995      0.935

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
Size
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━ 79/79 6.3it/s 12.5s
                   all       2526       2526      0.995      0.997
0.995      0.945
```

```
      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
Size
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━━━━━━━━━ 79/79 6.3it/s 12.6s
                   all       2526       2526      0.995      0.995
0.995      0.944

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
Size
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━━━━━━━━━ 79/79 6.3it/s 12.5s
                   all       2526       2526      0.994      0.997
0.995      0.942

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
Size
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━━━━━━━━━ 79/79 6.3it/s 12.6s
                   all       2526       2526      0.995      0.997
0.995      0.952

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
Size
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━━━━━━━━━ 79/79 6.3it/s 12.5s
                   all       2526       2526      0.994      0.997
0.995      0.952

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
Size
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━━━━━━━━━ 79/79 6.5it/s 12.2s
                   all       2526       2526      0.996      0.997
0.995      0.957

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
Size
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━━━━━━━━━ 79/79 6.4it/s 12.4s
                   all       2526       2526      0.995      0.997
0.995      0.954

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
Size
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━━━━━━━━━ 79/79 6.4it/s 12.4s
                   all       2526       2526      0.996      0.997
0.995      0.953
```

```
      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances
Size
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━━━━━━━━━ 79/79 6.3it/s 12.6s
                   all       2526       2526      0.996      0.997
0.995      0.956

30 epochs completed in 0.954 hours.
Optimizer stripped from /content/runs/detect/train/weights/last.pt,
18.5MB
Optimizer stripped from /content/runs/detect/train/weights/best.pt,
18.5MB

Validating /content/runs/detect/train/weights/best.pt...
Ultralytics 8.4.30 □ Python-3.12.13 torch-2.10.0+cu128 CUDA:0 (Tesla
T4, 14913MiB)
YOLOv5s summary (fused): 85 layers, 9,128,177 parameters, 0 gradients,
23.9 GFLOPs
ages  Instances      Box(P          R      mAP50  mAP50-95): 100%
━━━━━━━━━━━━━━━━━━━━ 79/79 5.4it/s 14.6s
                   all       2526       2526      0.996      0.997
0.995      0.957
                     0         10         10      0.989          1
0.995      0.913
                     1        151        151      0.998          1
0.995      0.956
                     2        150        150          1      0.993
0.995      0.972
                     3         84         84      0.999          1
0.995      0.961
                     4        151        151          1      0.982
0.995      0.953
                     5        130        130      0.999          1
0.995      0.967
                     6         29         29      0.998          1
0.995      0.937
                     7         82         82      0.999          1
0.995      0.932
                     8         85         85      0.999          1
0.995      0.952
                     9        102        102      0.999          1
0.995      0.956
                    10        133        133      0.999          1
0.995      0.952
                    11         88         88      0.999          1
0.995      0.969
                    12        114        114      0.999          1
0.995      0.965
                    13        158        158          1      0.995
```

0.995	0.978							
		14	52	52	0.998	1	0.995	0.974
		15	44	44	0.998	1	0.995	0.944
		16	28	28	0.998	1	0.995	0.964
		17	70	70	0.999	1	0.995	0.944
		18	84	84	0.999	1	0.995	0.96
		19	13	13	0.947	1	0.995	0.98
		20	20	20	1	0.956	0.995	0.91
		21	21	21	0.994	1	0.995	0.972
		22	21	21	0.997	1	0.995	0.971
		23	32	32	0.997	1	0.995	0.968
		24	21	21	0.996	1	0.995	0.98
		25	97	97	1	0.995	0.995	0.975
		26	29	29	0.997	1	0.995	0.963
		27	12	12	0.993	1	0.995	0.932
		28	34	34	0.998	1	0.995	0.962
		29	16	16	0.995	1	0.995	0.983
		30	35	35	0.997	1	0.995	0.936
		31	53	53	0.999	1	0.995	0.962
		32	14	14	0.993	1	0.995	0.931
		33	40	40	0.998	1	0.995	0.939
		34	15	15	0.996	1	0.995	0.953
		35	68	68	0.996	1	0.995	0.951
		36	23	23	0.996	1	0.995	0.953
		37	9	9	0.992	1	0.995	0.972

```
                    38        145        145          1          1
0.995       0.95
                    39         13         13      0.992          1
0.995      0.958
                    40         17         17      0.995      0.941
0.992      0.961
                    41          8          8       0.99          1
0.995      0.982
                    42         25         25      0.997          1
0.995      0.972
Speed: 0.1ms preprocess, 1.6ms inference, 0.0ms loss, 1.4ms
postprocess per image
Results saved to /content/runs/detect/train

ultralytics.utils.metrics.DetMetrics object with attributes:

ap_class_index: array([ 0,  1,  2,  3,  4,  5,  6,  7,  8,  9, 10, 11,
12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28,
29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42])
box: ultralytics.utils.metrics.Metric object
confusion_matrix: <ultralytics.utils.metrics.ConfusionMatrix object at
0x78dc2d43a330>
curves: ['Precision-Recall(B)', 'F1-Confidence(B)', 'Precision-
Confidence(B)', 'Recall-Confidence(B)']
curves_results: [[array([          0,    0.001001,    0.002002,
0.003003,    0.004004,    0.005005,    0.006006,    0.007007,
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0.018018,    0.019019,     0.02002,    0.021021,    0.022022,
0.023023,
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0,           0]]), 'Confidence', 'Recall']]
fitness: 0.9573227967302659
keys: ['metrics/precision(B)', 'metrics/recall(B)',
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maps: array([    0.91253,     0.95614,     0.97221,     0.96068,
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0.97431,     0.94378,     0.96378,     0.94447,     0.95974,
0.97991,     0.91007,     0.97187,      0.9713,     0.96843,
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0.96221,     0.98308,     0.93569,      0.9618,     0.93069,
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145,  13,  17,   8,  25])
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results_dict: {'metrics/precision(B)': 0.9959149015241507,
'metrics/recall(B)': 0.996814687545434, 'metrics/mAP50(B)':
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'fitness': 0.9573227967302659}
save_dir: PosixPath('/content/runs/detect/train')
speed: {'preprocess': 0.10297876880770804, 'inference':
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1.3906595387976894}
stats: {'tp': [], 'conf': [], 'pred_cls': [], 'target_cls': [],
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```

```
model = YOLO("runs/detect/train/weights/best.pt")

model("/content/dataset/images/val", save=True)
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